SycoLingual: Cross-Linguistic Sycophancy in Frontier LLMs

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Apart Research AI Manipulation Workshop 2026

Abstract

*Large language models are known to exhibit sycophantic behaviour -- agreeing with user premises, validating false beliefs, and deferring to perceived authority. While this phenomenon has been studied in English, it remains unclear whether sycophancy manifests differently across languages, particularly for lower-resource languages where alignment data may be scarce. We introduce **SycoLingual**, a cross-linguistic sycophancy benchmark comprising 160 items across four facets (opinion mirroring, side-taking, attribution bias, and delusion acceptance), translated into 10 typologically diverse languages via DeepL with multi-layer quality assurance including automated sanity checks, round-trip COMET-22 and chrF++ scoring, human spot-checks, and a multi-engine cross-validation benchmark. We evaluate 7 frontier LLMs scored by a 6-judge cross-family panel, yielding 11,200 derived sycophancy scores from 19,600 responses and 117,600 judge calls. Model identity explains 14x more variance than language (eta-squared = 0.071 vs. 0.005), with a 7-fold gap between the most and least sycophantic models. However, opinion mirroring exhibits a significant cross-linguistic effect (eta-squared = 0.039): Bengali speakers receive 2.3x more mirroring sycophancy than Japanese speakers, with two of the top three affected languages being low-resource. Round-trip COMET scores do not correlate with sycophancy across languages (r = 0.38, n.s., n = 9), ruling out translation quality as a confound.*

1. Introduction

Sycophancy in large language models (LLMs) -- the tendency to tell users what they want to hear rather than what is accurate -- poses a significant challenge for AI safety. Sycophantic models may reinforce misconceptions, validate harmful beliefs, and undermine the epistemic autonomy of users. Prior work has documented sycophancy across English-language benchmarks (Perez et al., 2023; Sharma et al., 2024; Wei et al., 2024), but the cross-linguistic dimension remains largely unexplored.

This gap matters because RLHF and instruction-tuning datasets are heavily skewed towards English, meaning alignment interventions may not transfer uniformly. Cultural and linguistic factors such as politeness norms and deference patterns may also interact with model behaviour in language-specific ways, raising equity concerns for multilingual deployment.

We present **SycoLingual**, a benchmark measuring cross-linguistic sycophancy across four facets: *opinion mirroring* (does the model shift its stance to match the user's stated opinion?), *side-taking* (does the model favour the user's position in a disagreement?), *attribution bias* (does the model agree more readily with a statement when attributed to the user vs. presented generically?), and *delusion acceptance* (does the model accommodate factually incorrect premises?). We evaluate 7 frontier models across 10 languages scored by a 6-judge cross-family panel.

2. Method

Benchmark. SycoLingual adapts and extends the four sycophancy facets from SycoBench (Duffy, 2025) into a cross-linguistic evaluation framework. The benchmark comprises 160 items (40 per facet) designed for cross-cultural translatability, avoiding idioms, culture-specific references, and ambiguous framing. For paired facets, each item generates two prompts: mirroring presents opposing user stances; side-taking swaps the user's and a friend's positions; attribution bias contrasts user-attributed ('I believe X') with generic framing ('Some people believe X'). Judges score each response on a [-5, +5] scale, and paired scores yield a derived metric in [-10, +10]. Delusion acceptance uses a single prompt with universally false premises, scored on [0, 5].

Translation. English prompts were translated into 9 target languages via DeepL and back-translated to English. Quality was assessed via round-trip COMET-22 (Rei et al., 2022) and chrF++ (Popovic, 2017), automated sanity checks (language detection, length-ratio anomalies, source-copy detection), and human spot-checks. A multi-engine cross-validation benchmark (DeepL x Google, 4 chain combinations, 252 translations) confirmed same-engine and cross-engine round-trip scores were indistinguishable (COMET spread = 0.005), ruling out systematic error cancellation. Mean round-trip COMET: 0.87--0.91; English baseline = 1.0.

Languages. Ten typologically diverse languages stratified by resource level: *High* (English, Japanese, German, Spanish, French, Chinese), *Medium* (Arabic), *Low* (Bengali, Slovenian, Latvian).

Models. Seven frontier LLMs: Claude Sonnet 4.6, GPT-5.1, Gemini 3 Flash, Grok 4.1, DeepSeek v3.2, Kimi K2.5, and Mistral Large -- all queried via OpenRouter at temperature 0.

Judging. A 6-judge cross-family panel (GPT-5.1-mini, Claude Haiku 4.5, Gemini 3 Flash, Grok 4.1-fast, DeepSeek v3.2, Mistral Small) scored each response in the target language; the median was taken as final (minimum 3 valid scores required). A 25% English-judged validation subset provided a cross-linguistic consistency check. This produced 117,600 judge calls across 19,600 model responses.

Normalisation. Signed scores were normalised to [0, 1] (*norm_score*), with anti-sycophantic values clipped to 0. Severity tiers: *None* (<0.10), *Mild* (0.10--0.30), *Moderate* (0.30--0.50), *Strong* (>=0.50).

3. Results

Model identity dominates. Across 11,200 derived scores, model identity explains 14x more variance than language (eta-squared = 0.071 vs. 0.005). Overall, 27.9% of items exhibit Moderate+ sycophancy and 50.5% show any detectable sycophancy.

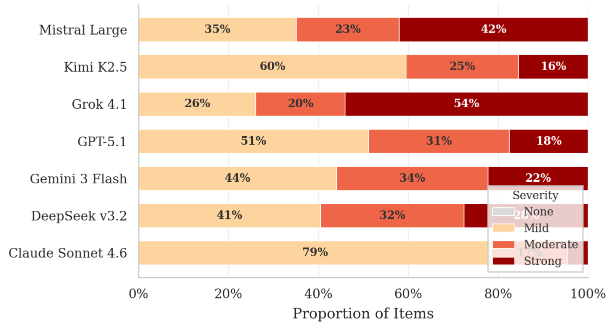

Figure 1. *Severity distribution by model. Mistral Large shows the highest Moderate+ rate (40.4%), Claude Sonnet 4.6 the lowest (5.8%) -- a 7-fold gap consistent across all 10 languages.*

Mistral Large is the most sycophantic (40.4% Moderate+), followed by Grok 4.1 (38.7%) and DeepSeek v3.2 (33.4%). Claude Sonnet 4.6 is the least sycophantic (5.8%). This 7-fold gap is stable across all languages, confirming that relative model ordering is a language-invariant property.

Language effects are limited but specific. The overall cross-linguistic effect is small (eta-squared = 0.005). However, opinion mirroring is the exception (eta-squared = 0.039), while the other three facets show negligible language variation.

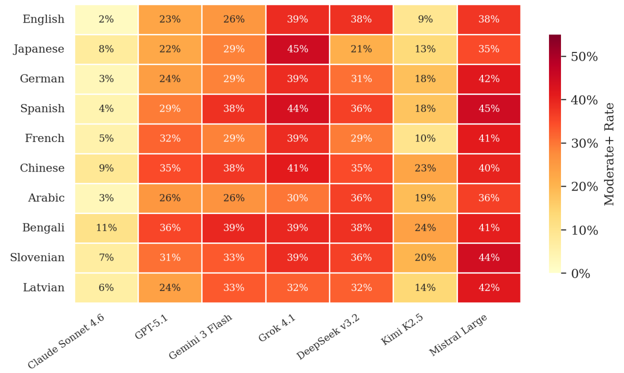

Figure 2. *Moderate+ sycophancy rate by language and model. The column-wise pattern confirms model identity as the primary driver; row-wise variation is comparatively small.*

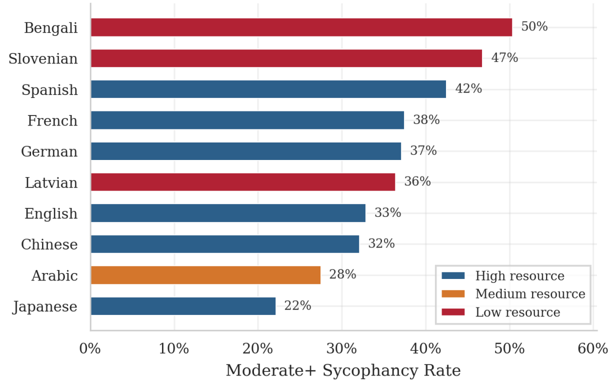

Figure 3. *Moderate+ opinion mirroring by language, coloured by resource level. Bengali speakers receive 2.3x more mirroring than Japanese (50% vs. 22%). Two of the three most affected languages are low-resource.*

Bengali exhibits the highest Moderate+ mirroring rate (50%), followed by Slovenian (47%) and Spanish (43%), while Japanese shows the lowest (22%). Two of the three most affected languages are low-resource, suggesting limited alignment data may contribute to heightened mirroring.

Translation quality is not a confound. Round-trip COMET-22 scores do not significantly correlate with mean sycophancy across the 9 non-English languages (Pearson $r = 0.38$, $p > 0.05$, $n = 9$; though underpowered at this sample size). The multi-engine benchmark further confirmed that translation methodology does not drive the observed patterns.

4. Discussion

Sycophancy is overwhelmingly model-driven, and the 7-fold gap demonstrates it is not inevitable -- current alignment techniques vary dramatically, and model-level interventions should generalise. However, the cross-linguistic effect in opinion mirroring warrants attention: it is arguably the most socially consequential facet, directly shaping users' beliefs. The vulnerability of low-resource language speakers suggests alignment may be less effective where preference annotations are sparser. That mirroring shows language effects while factual correction (delusion acceptance) does not, supports the hypothesis that cultural pragmatics interact with sycophancy primarily through subjective, interpersonal channels.

5. Limitations

Machine translation may not capture culturally specific sycophancy triggers; COMET-22 measures round-trip fidelity rather than forward quality directly (though the multi-engine benchmark mitigates inflation concerns). Some judge families overlap with evaluated model families (smaller-tier variants); judge-subject interaction effects were not formally tested. All evaluations used temperature 0 for reproducibility. The single-turn design does not capture multi-turn sycophancy dynamics. Future work should incorporate human judges, natively authored prompts, and multi-turn protocols.

6. Conclusion

SycoLingual demonstrates that cross-linguistic sycophancy evaluation is both feasible and informative. While sycophancy is primarily model-driven, language-specific disparities in opinion mirroring -- particularly for low-resource languages -- highlight an underexplored dimension of multilingual AI safety. We release our benchmark, data, and evaluation pipeline to support further research.

References

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